



GE Medical Systems

Technical Publications

Direction 2352789
Revision 0

Signa[®] RFI Cable Kit Installation

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Operating Documentation

DAMAGE IN TRANSPORTATION

All packages should be closely examined at time of delivery. If damage is apparent, have notation “**damage in shipment**” written on **all** copies of the freight or express bill **before** delivery is accepted or “signed for” by a General Electric representative or a hospital receiving agent. Whether noted or concealed, damage **MUST** be reported to the carrier **immediately** upon discovery, or in any event, within **14** days after receipt, and the contents and containers held for inspection by the carrier. A transportation company will not pay a claim for damage if an inspection is not requested within this **14** day period.

Immediately complete a "Damage Loss Claim Form", available via MS Exchange Mail, after the damage is found.

MS Exchange Path:

Outlook/Public Folder/All Public Folders/Medical Systems/!Global Initiatives/Information Management/Forms/Common Forms/DAMAGE LOSS CLAIM FORM.

Send the completed form to the email address listed in the form.

For more information about the Transportation Claim Procedure, access the GE Medical Systems Intranet and enter the following URL address (case sensitive):

<ftp://3.87.40.2/globepro/qualsys/Docs/190016MF.PDF>

Rev. 11/15/2000

Direction 2128126 - Language Policy For Service Documentation, Rev 1

WARNING

- THIS SERVICE MANUAL IS AVAILABLE IN ENGLISH ONLY.
- IF A CUSTOMER'S SERVICE PROVIDER REQUIRES A LANGUAGE OTHER THAN ENGLISH, IT IS THE CUSTOMER'S RESPONSIBILITY TO PROVIDE TRANSLATION SERVICES.
- DO NOT ATTEMPT TO SERVICE THE EQUIPMENT UNLESS THIS SERVICE MANUAL HAS BEEN CONSULTED AND IS UNDERSTOOD.
- FAILURE TO HEED THIS WARNING MAY RESULT IN INJURY TO THE SERVICE PROVIDER, OPERATOR OR PATIENT FROM ELECTRIC SHOCK, MECHANICAL OR OTHER HAZARDS.

AVERTISSEMENT

- CE MANUEL DE MAINTENANCE N'EST DISPONIBLE QU'EN ANGLAIS.
- SI LE TECHNICIEN DU CLIENT A BESOIN DE CE MANUEL DANS UNE AUTRE LANGUE QUE L'ANGLAIS, C'EST AU CLIENT QU'IL INCOMBE DE LE FAIRE TRADUIRE.
- NE PAS TENTER D'INTERVENTION SUR LES ÉQUIPEMENTS TANT QUE LE MANUEL SERVICE N'A PAS ÉTÉ CONSULTÉ ET COMPRIS.
- LE NON-RESPECT DE CET AVERTISSEMENT PEUT ENTRAÎNER CHEZ LE TECHNICIEN, L'OPÉRATEUR OU LE PATIENT DES BLESSURES DUES À DES DANGERS ÉLECTRIQUES, MÉCANIQUES OU AUTRES.

WARNUNG

- DIESES KUNDENDIENST-HANDBUCH EXISTIERT NUR IN ENGLISCHER SPRACHE.
- FALLS EIN FREMDER KUNDENDIENST EINE ANDERE SPRACHE BENÖTIGT, IST ES AUFGABE DES KUNDEN FÜR EINE ENTSPRECHENDE ÜBERSETZUNG ZU SORGEN.
- VERSUCHEN SIE NICHT, DAS GERÄT ZU REPARIEREN, BEVOR DIESES KUNDENDIENST-HANDBUCH NICHT ZU RATE GEZOGEN UND VERSTANDEN WURDE.
- WIRD DIESE WARNUNG BEACHTET, SO KANN ES ZU VERLETZUNGEN DES KUNDENDIENSTTECHNIKERS, DES BEDIENERS ODER DES PATIENTEN DURCH ELEKTRISCHE SCHLÄGE, MECHANISCHE ODER SONSTIGE GEFAHREN KOMMEN.

AVISO

- ESTE MANUAL DE SERVICIO SÓLO EXISTE EN INGLÉS
- SI ALGÚN PROVEEDOR DE SERVICIOS AJENO A GEMS SOLICITA UN IDIOMA QUE NO SEA EL INGLÉS, ES RESPONSABILIDAD DEL CLIENTE OFRECER UN SERVICIO DE TRADUCCIÓN.
- NO SE DEBERÁ DAR SERVICIO TÉCNICO AL EQUIPO, SIN HABER CONSULTADO Y COMPRENDIDO ESTE MANUAL DE SERVICIO.
- LA NO OBSERVANCIA DEL PRESENTE AVISO PUEDE DAR LUGAR A QUE EL PROVEEDOR DE SERVICIOS, EL OPERADOR O EL PACIENTE SUFRAN LESIONES PROVOCADAS POR CAUSAS ELÉCTRICAS, MECÁNICAS O DE OTRA NATURALEZA.

AVVERTENZA

- IL PRESENTE MANUALE DI MANUTENZIONE È DISPONIBILE SOLTANTO IN INGLESE.
- SE UN ADDETTO ALLA MANUTENZIONE ESTERNO ALLA GEMS RICHIEDE IL MANUALE IN UNA LINGUA DIVERSA, IL CLIENTE È TENUTO A PROVVEDERE DIRETTAMENTE ALLA TRADUZIONE.
- SI PROCEDA ALLA MANUTENZIONE DELL'APPARECCHIATURA SOLO DOPO AVER CONSULTATO IL PRESENTE MANUALE ED AVERNE COMPRESO IL CONTENUTO.
- NON TENERE CONTO DELLA PRESENTE AVVERTENZA POTREBBE FAR COMPIERE OPERAZIONI DA CUI DERIVINO LESIONI ALL'ADDETTO ALLA MANUTENZIONE, ALL'UTILIZZATORE ED AL PAZIENTE PER FOLGORAZIONE ELETTRICA, PER URTI MECCANICI OD ALTRI RISCHI.

警告

- ・このサービスマニュアルは 英語版しかありません。
- ・GEMS以外でサービスを担当される業者が英語以外の言語を要求される場合、翻訳作業はその業者の責任で行うものとさせていただきます。
- ・このサービスマニュアルを熟読し、理解せずに装置のサービスを行わないでください。
- ・この警告に従わない場合、サービスを担当される方、操作員あるいは患者さんが、感電や機械的又はその他の危険により負傷する可能性があります。

注意:

- 本维修手册仅存有英文本。
- 非 GEMS 公司的维修员要求非英文本的维修手册时，客户需自行负责翻译。
- 未详细阅读和完全了解本手册之前，不得进行维修。
- 忽略本注意事项会对维修员，操作员或病人造成触电，机械伤害或其他伤害。

REVISION HISTORY

<u>REV</u>	<u>DATE</u>	<u>PRIMARY REASON FOR CHANGE</u>
0	Aug. 22, 2002	Initial version of RFI Cable installation procedure.

<u>PAGE</u>	<u>REV</u>	<u>PAGE</u>	<u>REV</u>	<u>PAGE</u>	<u>REV</u>	<u>PAGE</u>	<u>REV</u>
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* This revision/letter corresponds to the indicated document's revision control system.

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1- OVERVIEW

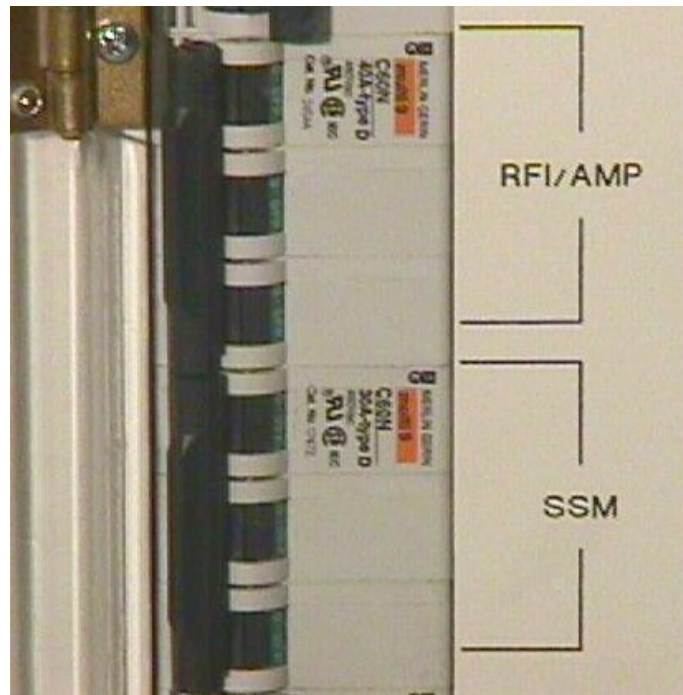
The RFI Cable Kit, part number 2349501, consists of two shorter cables for connection of RF Out (J4) from each amplifier to the RFI (J1 & J2) to address the loss of amplifier performance due to mismatch.

2 – SRF CABINET POWER LOCKOUT / TAGOUT



POSSIBLE PERSONAL INJURY! AVOID SERIOUS INJURY OR DEATH BY ELECTROCUTION. REMOVE POWER FROM THE SRF CABINET BEFORE ATTEMPTING TO REMOVE THE RF AMPLIFIER. VERIFY THAT THE RFI/AMP AND SSM BREAKERS ON THE FRONT OF THE PDU ARE OFF. LOCK AND TAG OUT THE PLEXI-GLASS COVER ON THE FRONT OF THE PDU.

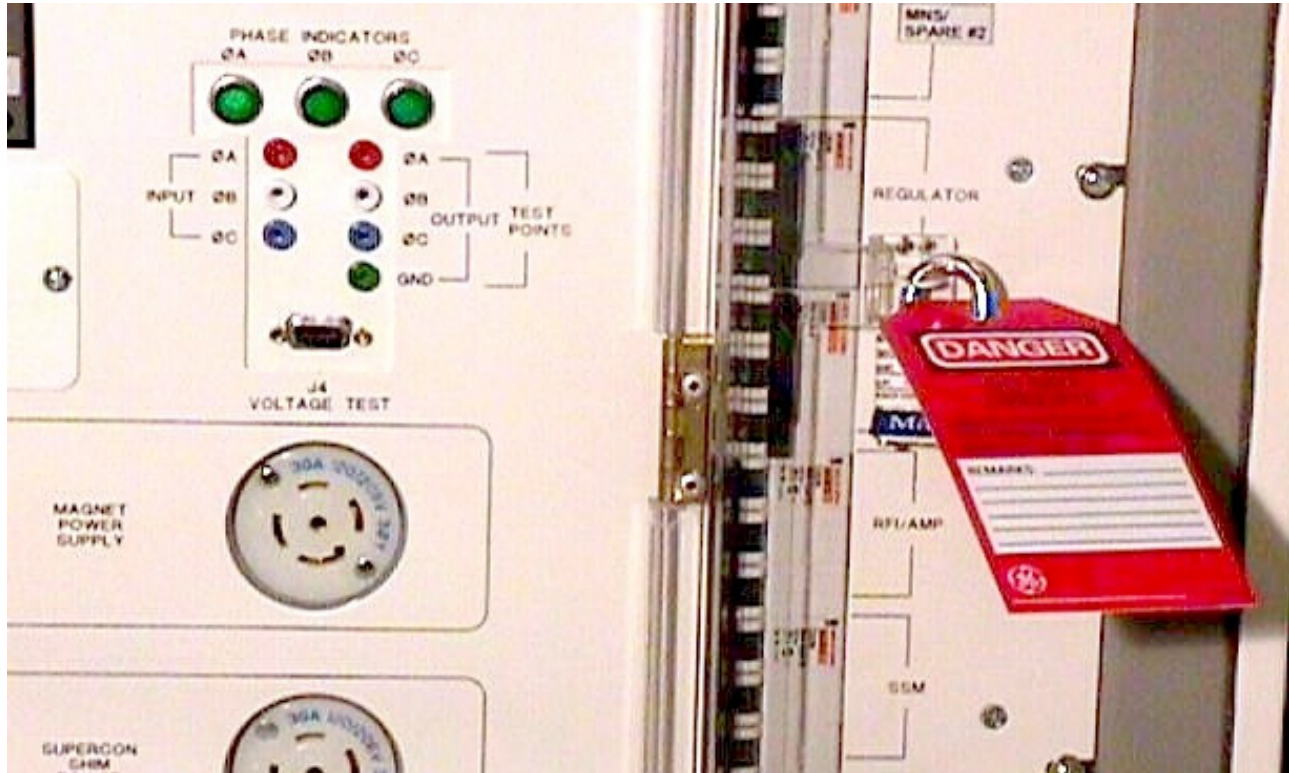
1. Remove the cover from the front of the ACGD cabinet.
2. Open Plexiglas door over the front of the PDU breakers and switch off the **RFI/AMP** and **SSM** breakers. See Illustration 2-1.



BREAKER IDENTIFICATION
ILLUSTRATION 2-1

2 – SRF CABINET POWER LOCKOUT AND TAGOUT (CONTINUED)

3. Close the plexi-glass door and lock and tag it out. See Illustration 2-2. (Refer to CD-ROM *Dir. 2187583-3 [or -2], MR Release Signa 5x/8x Service Methods, Renewal Parts and Service Tools, Safety, Section 6, OSHA LOCKOUT/TAGOUT REQUIREMENTS.*)



LOCKOUT AND TAGOUT OF BREAKERS
ILLUSTRATION 2-2

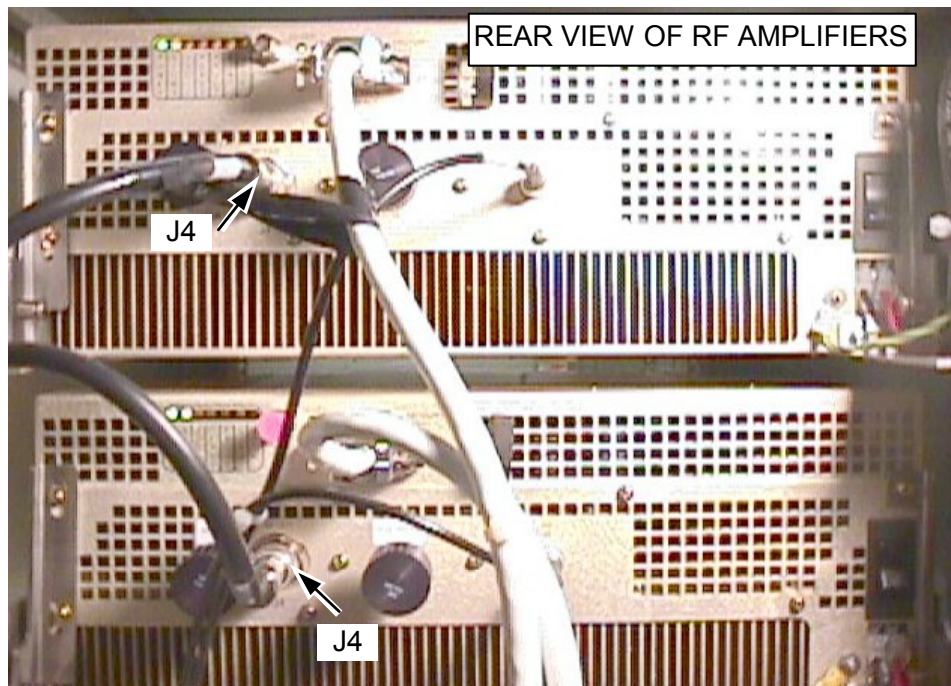
4. Remove the front cover from the SRF Cabinet.
5. Verify all LEDs are OFF (not illuminated) on front of SSM and RFI. Verify all LEDs are OFF (not illuminated) at the rear of the RF Amplifier Power Module.
6. Using an AC voltmeter (DVM), measure the input power leads on the rear of one of the RF Amplifier Modules from the rear (right side) of the SRF Cabinet.
 - Measure between L1 and L2; verify voltmeter measures 0 VAC.

3 - REMOVAL/REPLACEMENT OF RFI CABLES

DANGER!!

POSSIBLE PERSONAL INJURY! AVOID SERIOUS INJURY OR DEATH BY ELECTROCUTION. VERIFY THAT LOCKOUT / TAGOUT OF THE SSM AND RFI/AMP CIRCUIT BREAKERS ON THE FRONT OF THE PHOENIX PDU HAVE BEEN PROPERLY PERFORMED.

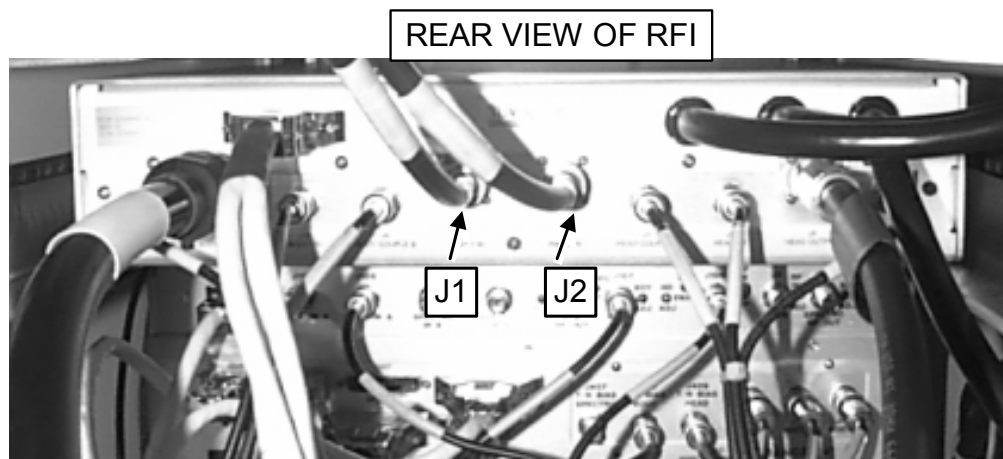
1. Perform Section 2 – **SRF CABINET POWER LOCKOUT AND TAGOUT** (Refer to 8.X Service Methods CD-ROM, Safety (SYSMSA2.DOC), Section 6, OSHA LOCKOUT/TAGOUT REQUIREMENTS.)
2. Remove cable connecting to J4 on both RF amplifiers. See Illustration 3-1.



J4 LOCATION ON RF AMPLIFIERS
ILLUSTRATION 3-1

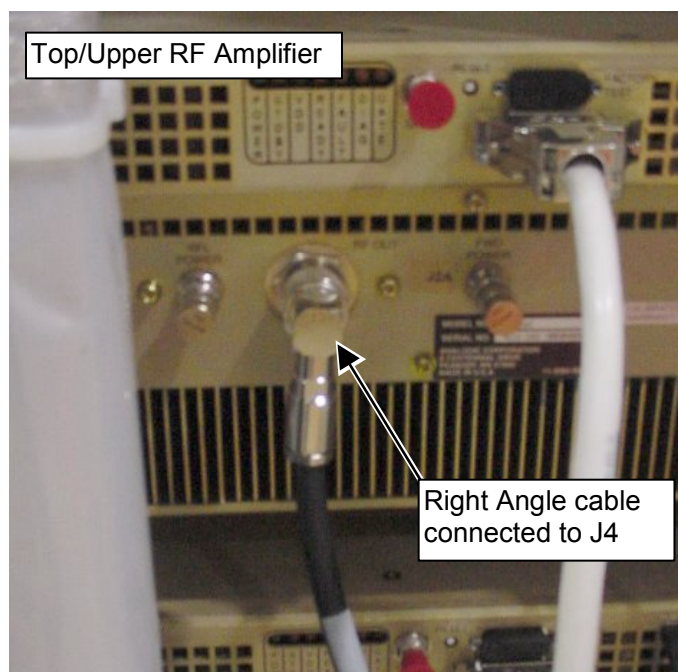
3 - REMOVAL/REPLACEMENT OF RFI CABLES (continued)

3. Remove J1 and J2 connections from rear of RFI. See Illustration 3-2.



LOCATION OF J1 & J2 ON REAR OF RFI MODULE
ILLUSTRATION 3-2

4. Connect the new right angle connector cable to J4 of the top/upper amplifier in the cabinet. See Illustration 3-3.



RIGHT ANGLE CABLE CONNECTED TO J4
ILLUSTRATION 3-3

3 - REMOVAL/REPLACEMENT OF RFI CABLES (continued)

5. Connect other end of cable to J1 on the rear of the RFI module. See Illustration 3-4.



NEW RIGHT ANGLE CABLE INSTALLED
ILLUSTRATION 3-4

3 - REMOVAL/REPLACEMENT OF RFI CABLES (continued)

6. Connect the other right angle connector to J4 of the lower amplifier. See Illustration 3-5.

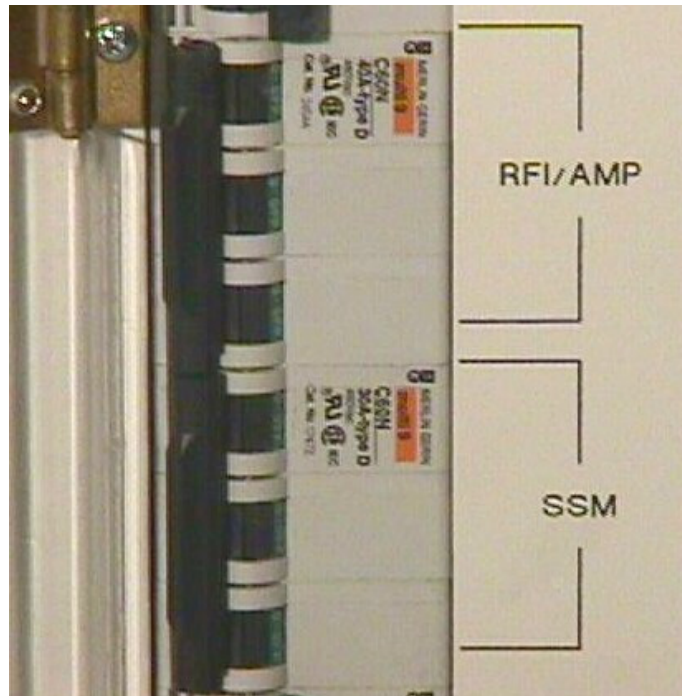


RIGHT ANGLE CABLE CONNECTED TO LOWER RF AMP
ILLUSTRATION 3-5

7. Connect other end of cable to J2 of the RFI module. See Illustration 3-5.

4 – SYSTEM RESTORATION

1. Open Plexiglas door over the front of the PDU breakers and switch ON the **RFI/AMP** and **SSM** breakers. See Illustration 4-1.



BREAKER IDENTIFICATION
ILLUSTRATION 4-1

2. Verify all LEDs are ON (illuminated) on front of SSM and RFI. Verify all LEDs are ON (illuminated) at the rear of the RF Amplifier Power Module.
3. Replace the cover on the ACGD Cabinet.
4. Perform the balance & gain adjustments in *1.5T SRF Max Power RF Out Setup Calibration (RC5SCA1)* procedure.